

Principles of Internal Medicine and Clinical Examination

Internal Medicine — MBBS 3rd to Final Year

Standard Core Curriculum for Medical Undergraduates • Comprehensive Clinical Reference

Curriculum Year	MBBS 3rd to Final Year
Medical Subject	Internal Medicine
Target Audience	MBBS Undergraduates, Postgraduates, USMLE/PLAB Aspirants
Format	Fully Indexed Textbook with Chapter, Topic, and Page Referencing
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Chapter 1: Cardiology and Acute Coronary Syndromes
Acute Chest Pain Workup, STEMI vs NSTEMI, and Immediate Management

Chest pain requires rapid risk stratification to rule out life-threatening causes: acute coronary syndrome (ACS), aortic dissection, pulmonary embolism, tension pneumothorax, esophageal rupture, and cardiac tamponade.

An ECG must be acquired within 10 minutes. STEMI is defined as new ST-segment elevation at the J-point in >= 2 contiguous leads (> 1 mm in all leads except V2-V3, where thresholds are 1.5-2.5 mm) or new LBBB. Positive troponin without ST elevation indicates NSTEMI.

Immediate emergency management of STEMI is rapid reperfusion: Primary PCI within 90 minutes door-to-balloon time. If PCI unavailable within 120 minutes, intravenous fibrinolytic therapy (Tenecteplase) within 30 minutes door-to-needle time. Adjunctive therapy includes dual antiplatelets (Aspirin 300mg + Ticagrelor 180mg), anticoagulation (Heparin/Enoxaparin), high-intensity statin (Atorvastatin 80mg), and sublingual nitroglycerin (avoided in RV infarction).

Clinical Pearl: Avoid nitrates and morphine in Right Ventricular Infarction (inferior ST elevations in II, III, aVF with ST elevation in V4R) because RV output is critically preload-dependent; nitrates precipitate severe shock.

	Troponin Biomarkers	Reperfusion Strategy
	Markedly elevated	Immediate Primary PCI
	Elevated above 99th percentile	Dual antiplatelet therapy + P2Y12 inhibitor
	Normal (no myocardial necrosis)	Antiplatelet, anticoagulation

Chapter 1: Cardiology and Acute Coronary Syndromes

Heart Failure Classification, NYHA Stages and Guideline Medical Therapy (GDMT)

Heart Failure (HF) is a clinical syndrome of cardinal symptoms (dyspnea, orthopnea, paroxysmal nocturnal dyspnea, fatigue, peripheral edema) from structural or functional abnormalities that impair ventricular filling or ejection.

HF is categorized by Left Ventricular Ejection Fraction (LVEF): (1) HFrEF (LVEF <= 40%); (2) HFmrEF (LVEF 41-49%); (3) HFpEF (LVEF >= 50%). Functional severity is graded by NYHA Class I (no limitation) to Class IV (symptoms at rest).

Guideline-directed medical therapy (GDMT) for HFrEF consists of the 'Four Pillars' proven to reduce mortality: (1) SGLT2 Inhibitors (Dapagliflozin/Empagliflozin); (2) ARNI (Sacubitril/Valsartan) or ACEi/ARB; (3) Beta-blockers (Carvedilol, Metoprolol succinate, Bisoprolol); (4) Mineralocorticoid Receptor Antagonists (MRA: Spironolactone/Eplerenone). Loop diuretics (Furosemide) provide symptom relief without mortality benefit.

Clinical Pearl: The 4 Pillars of HFrEF mortality reduction: ARNI/ACEi + Beta-blocker + MRA (Spironolactone) + SGLT2 inhibitor (Dapagliflozin). Furosemide provides symptom control but does not reduce mortality.

	Mechanistic Target	Proven Benefit
	Neprilysin inhibition + AT1 blockade	20% mortality reduction
	Anti-sympathetic neurohormonal blockade	34% mortality reduction
	Aldosterone antagonism in distal nephron	30% mortality reduction
	Inhibition of renal SGLT2; cardiac metabolic switch	Reduces CV death

Chapter 2: Pulmonology and Nephrology

Community-Acquired Pneumonia: CURB-65 Severity Scoring and Antimicrobial Regimens

Community-Acquired Pneumonia (CAP) presents with fever, cough, purulent sputum, pleuritic pain, and new focal infiltrate on chest radiography. *Streptococcus pneumoniae* remains the most frequent pathogen.

Triage is guided by CURB-65 score: (1) Confusion; (2) Urea > 7.0 mmol/L; (3) Respiratory rate ≥ 30 /min; (4) Blood pressure low (SBP < 90 or DBP ≤ 60 mmHg); (5) Age ≥ 65 . Score 0-1 = low risk (outpatient); Score 2 = moderate risk (inpatient); Score 3-5 = high risk (urgent hospital/ICU admission).

Empiric antibiotics: Outpatient low-risk receives Amoxicillin, Doxycycline, or Azithromycin. Hospitalized non-severe receives Ceftriaxone plus Azithromycin or respiratory fluoroquinolone monotherapy (Levofloxacin). Severe ICU CAP receives IV Ceftriaxone + Azithromycin + Vancomycin/Linezolid if MRSA suspected.

Clinical Pearl: CURB-65 criteria: Confusion, Urea > 7 mmol/L, Resp rate ≥ 30 , BP < 90/60, Age ≥ 65 . Score ≥ 3 requires urgent inpatient/ICU hospitalization.

	30-Day Mortality	Recommendation
	0.7% to 2.1%	Outpatient oral
	9.2%	Inpatient medical
	15% to 40%	Urgent inpatient/ICU

Chapter 2: Pulmonology and Nephrology

Acute Kidney Injury (AKI): KDIGO Staging, Diagnostic Workup and Management

Acute Kidney Injury (AKI) is defined by KDIGO criteria: increase in serum creatinine ≥ 0.3 mg/dL within 48h, ≥ 1.5 x baseline within 7 days, or urine output < 0.5 mL/kg/h for 6 hours.

AKI is categorized into: (1) Prerenal Azotemia: renal hypoperfusion without structural damage (dehydration, sepsis, heart failure, ACEi/NSAIDs), characterized by high urine osmolality (>500 mOsm/kg), fractional excretion of sodium $\text{FeNa} < 1\%$, and BUN/Cr ratio $> 20:1$; (2) Intrinsic AKI: structural parenchymal damage. Acute Tubular Necrosis (ATN) is most common (ischemia, aminoglycosides, IV contrast), showing $\text{FeNa} > 2\%$ and 'muddy brown' granular casts; (3) Postrenal AKI: obstruction (BPH, calculi), diagnosed by hydronephrosis on ultrasound.

Management involves treating the reversible cause: rapid IV crystalloids in prerenal hypovolemia, discontinuing nephrotoxic drugs, and urgent renal replacement therapy indications (AEIOU: Acidosis, Electrolytes/Hyperkalemia, Ingestion/Toxins, Overload/Pulmonary edema, Uremic pericarditis/encephalopathy).

Clinical Pearl: Prerenal AKI has $\text{FeNa} < 1\%$ and $\text{BUN/Cr} > 20:1$ with bland urine; Intrinsic ATN has $\text{FeNa} > 2\%$ and classic 'muddy brown' granular casts. Dialysis indications = AEIOU.

	Intrinsic AKI (Acute Tubular Necrosis)	Postrenal Obstruction
	$< 15 : 1$	Variable (early $> 1.0\%$ to 2.0%)
	$> 2.0\%$	$> 1.0\%$ to 2.0%
	< 350 mOsm/kg (isosthenuric)	< 400 mOsm/kg
	'Muddy brown' coarse granular casts, epithelial casts	RBCs, crystals,

Chapter 3: Infectious Diseases and Clinical Neuro-Infections

Acute Bacterial, Viral and Tuberculous Meningitis: Clinical Triad, CSF Differentiation, and Empiric Regimens

Meningitis is an acute inflammation of the leptomeninges (pia and arachnoid mater) and the subarachnoid space. Bacterial etiology varies by age: *Streptococcus pneumoniae* is the most common cause in adults and children over 2 months; *Neisseria meningitidis* is frequent in adolescents and young adults (associated with petechial/purpuric skin lesions in meningococemia); *Listeria monocytogenes* occurs in neonates, adults > 50 years, and immunocompromised hosts; Group B *Streptococcus* (*S. agalactiae*) and *E. coli* predominate in neonates. Viral (aseptic) meningitis is most often caused by Enteroviruses.

The classic clinical triad comprises High Fever, Headache with photophobia, and Nuchal Rigidity (neck stiffness), present in >70% of cases. Physical examination reveals positive meningeal signs: Kernig's sign (resistance or pain upon passive knee extension when the patient is supine with hip flexed 90°) and Brudzinski's sign (involuntary flexion of hips and knees elicited by passive neck flexion). In meningococcal infection, rapidly progressing petechiae or purpura fulminans may be present. A non-contrast head CT should precede Lumbar Puncture (LP) ONLY if risk of herniation exists (papilledema, new-onset seizures, focal neurological deficits, altered consciousness GCS < 10, or immunocompromise).

Emergency Management: 'Door-to-antibiotic' time must be under 1 hour; antibiotics must NEVER be delayed for CT scanning or LP. Empiric therapy for adults aged 3 months to 50 years is Ceftriaxone (2g IV q12h) PLUS Vancomycin (15-20 mg/kg IV q8-12h). In patients > 50 years or immunocompromised, IV Ampicillin (2g IV q4h) MUST be added to cover *Listeria*. Adjunctive Dexamethasone (10 mg IV q6h) must be administered with or 15 minutes before the first dose of antibiotics to decrease mortality and sensorineural hearing loss in *S. pneumoniae*. Close contacts of *N. meningitidis* require chemoprophylaxis with Rifampin (600 mg BID x 2d) or Ciprofloxacin (500 mg single dose).

Clinical Pearl: Never delay empiric antibiotics for CT or lumbar puncture in suspected acute bacterial meningitis. Dexamethasone (10 mg IV) must precede or accompany the first antibiotic dose to prevent neurological sequelae. Add Ampicillin for age > 50 to cover *Listeria*.

Acute Bacterial Meningitis	Viral (Aseptic) Meningitis
Markedly Elevated (> 250 mm H2O)	Normal to Mildly Elevated
Markedly Elevated (> 1000 - 10,000 / uL)	Elevated (50 - 500 / uL)
Marked Neutrophils (PMNs >= 80%)	Lymphocytes (Mononuclear >= 80%)
Markedly Elevated (> 100 - 500 mg/dL)	Normal to Slightly High (< 100 mg/dL)
Markedly Decreased (< 0.4 or < 40 mg/dL)	Normal (> 0.6; normal glucose)
Positive in 60-85% of cases	Negative (PCR detects Enterovirus)

Chapter 3: Infectious Diseases and Clinical Neuro-Infections

Pulmonary and Extrapulmonary Tuberculosis: Pathogenesis, Diagnostic Workup, and ATT (HRZE) Regimen

Tuberculosis (TB) is caused by Mycobacterium tuberculosis, an acid-fast bacillus transmitted via airborne droplet nuclei. Primary pulmonary infection establishes a subpleural Ghon focus with regional hilar lymphadenopathy (Ghon complex), typically healing into a calcified Ranke complex.

Post-primary (reactivation) TB involves the apical-posterior segments of upper lung lobes due to high oxygen tension, producing productive cough, hemoptysis, low-grade evening fever, night sweats, and weight loss. Extrapulmonary TB accounts for 15-20% of cases, including TB lymphadenitis (scrofula), pleural effusion, skeletal TB (Pott's spine), and Tuberculous Meningitis.

Diagnosis is confirmed by sputum Ziehl-Neelsen acid-fast staining and rapid molecular testing with CBNAAT / GeneXpert MTB/RIF (detecting M. tuberculosis and rifampin resistance within 2 hours). First-line standard Anti-Tubercular Therapy (ATT) comprises a 2-month intensive phase of four drugs—Isoniazid (H), Rifampin (R), Pyrazinamide (Z), and Ethambutol (E)—followed by a 4-month continuation phase of Isoniazid and Rifampin (2HRZE / 4HR).

Clinical Pearl: Standard ATT protocol: 2 months of HRZE followed by 4 months of HR. Pyridoxine (Vitamin B6, 10-25 mg/day) must be co-administered with Isoniazid to prevent peripheral neuropathy.

	Key Toxicities & Adverse Effects	Clinical Monitoring
	Peripheral neuropathy (prevent with Pyridoxine/B6), Hepatitis, SLE-like syndrome	Baseline and periodic
	Orange-red discoloration of body fluids, Hepatitis, Potent CYP3A4 inducer	Warn patients of
ne	Hepatotoxicity (most potent among ATT), Hyperuricemia (gouty arthralgia)	Serum uric acid
esis	Retrobulbar optic neuritis (loss of red-green color discrimination, decreased acuity)	Visual acuity and

Chapter 4: Endocrinology, Gastroenterology and Acute Neurology

Diabetic Ketoacidosis (DKA) vs HHS: Pathophysiology, Fluid Protocols and Insulin Titration

Diabetic Ketoacidosis (DKA) is an acute, life-threatening metabolic complication of absolute or relative insulin deficiency combined with counter-regulatory hormone excess (glucagon, cortisol, catecholamines, growth hormone), most common in Type 1 Diabetes Mellitus. Accelerated lipolysis releases free fatty acids converted by liver beta-oxidation into ketone bodies (acetoacetate, beta-hydroxybutyrate), generating severe high anion gap metabolic acidosis (anion gap > 12).

Diagnostic triad: Hyperglycemia (blood glucose > 250 mg/dL), metabolic acidosis (arterial pH < 7.30, serum bicarbonate < 18 mEq/L), and ketonemia/ketonuria with positive beta-hydroxybutyrate. Patients present with Kussmaul respirations (rapid, deep breathing to exhale CO₂), fruity acetone breath, abdominal pain, nausea/vomiting, and severe dehydration (average 5-7 L fluid deficit). In contrast, Hyperosmolar Hyperglycemic State (HHS) features profound hyperglycemia (> 600 mg/dL), severe hyperosmolality (> 320 mOsm/kg), and severe dehydration (> 9 L deficit) without significant ketoacidosis.

Emergency Management: (1) Fluid resuscitation is the first priority: 0.9% Normal Saline at 1000 mL/hour during the first hour; (2) Potassium replacement: Insulin drives potassium intracellularly. If serum potassium is < 3.3 mEq/L, HOLD insulin and infuse IV KCl until K⁺ > 3.3 mEq/L to prevent fatal arrhythmias; (3) Intravenous regular insulin: 0.1 units/kg bolus followed by 0.1 units/kg/hour continuous infusion; (4) Add 5% Dextrose to fluids when blood glucose drops below 200 mg/dL to permit continued insulin infusion until the anion gap closes (bicarbonate ≥ 18 mEq/L, pH > 7.30).

Clinical Pearl: Never start insulin in DKA if serum potassium is below 3.3 mEq/L; replace potassium first to prevent fatal cardiac arrest. Add 5% dextrose when glucose reaches 200 mg/dL while continuing insulin until the anion gap is closed.

Diabetic Ketoacidosis (DKA)	Hyperosmolar Hyperglycemic State (HHS)
> 250 mg/dL (typically 300 - 600 mg/dL)	Markedly elevated (> 600 to 1200 mg/dL)
< 7.30 (severe if < 7.00)	> 7.30 (normal to mildly reduced)
< 18 mEq/L (severe if < 10 mEq/L)	> 18 mEq/L (near normal)
Markedly elevated (strongly positive)	Negative or trace
< 320 mOsm/kg	Markedly elevated (> 320 mOsm/kg)
> 12 mEq/L	Normal to slightly elevated (< 12 mEq/L)
< 7 Liters (~100 mL/kg)	9 to 12 Liters (~150 mL/kg)

Chapter 4: Endocrinology, Gastroenterology and Acute Neurology

Acute Ischemic Stroke vs Intracranial Hemorrhage: NIHSS, Non-Contrast CT, and Reperfusion Therapy

Stroke is an acute focal neurological deficit of vascular origin lasting > 24 hours. It is divided into Acute Ischemic Stroke (85%, caused by thrombosis or cardioembolism) and Intracranial Hemorrhage (15%, intracerebral or subarachnoid).

Emergency Triage: The immediate mandatory investigation is an urgent Non-Contrast Head CT (NCCT) to rule out intracranial hemorrhage before administering thrombolytic therapy. Ischemic stroke appears normal in the early hours (except early signs: hyperdense MCA sign, insular ribbon loss, sulcal effacement), whereas acute hemorrhage appears hyperdense (bright white).

Reperfusion Windows: For acute ischemic stroke, intravenous thrombolysis with Alteplase (0.9 mg/kg, max 90 mg) or Tenecteplase is recommended within 4.5 hours of symptom onset (target door-to-needle time < 45 min), provided blood pressure is < 185/110 mmHg and contraindications (active bleeding, recent surgery, platelets < 100k, INR > 1.7) are absent. Mechanical thrombectomy is indicated for Large Vessel Occlusion (LVO in internal carotid or proximal MCA) within 6 to 24 hours.

Clinical Pearl: Non-contrast head CT is the first test in acute stroke to rule out hemorrhage. The therapeutic window for IV thrombolysis (Alteplase) is 4.5 hours; mechanical thrombectomy extends up to 24 hours for large vessel occlusion.

	Primary Etiology	Acute Reperfusion Therapy
	Atherosclerotic plaque rupture / in-situ thrombosis	IV Alteplase within 4.5 hours
	Atrial fibrillation (left atrial appendage thrombus), endocarditis	IV Alteplase within 4.5 hours
	Chronic hypertension (Charcot-Bouchard aneurysms), amyloid angiopathy	BP lowering (target < 185/110 mmHg)
	Rupture of saccular berry aneurysm (circle of Willis)	Nimodipine (prevention of vasospasm)

Chapter 4: Endocrinology, Gastroenterology and Acute Neurology

Liver Cirrhosis, Portal Hypertension, and Esophageal Variceal Hemorrhage Management

Cirrhosis represents end-stage diffuse hepatic fibrosis with regenerative nodule formation, disrupting hepatic architecture and vascular flow. Chronic hepatitis B/C and alcohol are leading etiologies. Fibrous septa are deposited by activated Hepatic Stellate Cells (Ito cells) under TGF-beta stimulation.

Portal Hypertension (hepatic venous pressure gradient HVPG > 5 mmHg; clinically significant if >= 10 mmHg) produces cardinal clinical sequelae: Splenomegaly (hypersplenism and thrombocytopenia), Ascites, Portosystemic shunts (esophageal and gastric varices, caput medusae, hemorrhoids), and Hepatic Encephalopathy (elevated ammonia). Ascites workup requires diagnostic paracentesis to calculate the Serum-Ascites Albumin Gradient (SAAG): SAAG >= 1.1 g/dL indicates portal hypertension (cirrhosis, heart failure, Budd-Chiari); SAAG < 1.1 g/dL indicates peritoneal causes (carcinomatosis, tuberculosis, nephrotic syndrome).

Acute Esophageal Variceal Hemorrhage is a medical emergency: (1) Restrictive blood transfusion targeting hemoglobin 7-8 g/dL; (2) Vasoactive splanchnic vasoconstrictor: IV Octreotide (50 mcg bolus then 50 mcg/h) or Terlipressin; (3) Prophylactic antibiotic: Ceftriaxone (1g/day IV x 7 days) to prevent spontaneous bacterial peritonitis (SBP); (4) Emergency upper GI endoscopy within 12 hours for Endoscopic Variceal Ligation (EVL); (5) Secondary prophylaxis: non-selective beta-blockers (Propranolol, Carvedilol) plus repeated EVL until varices are eradicated.

Clinical Pearl: SAAG >= 1.1 g/dL confirms portal hypertension as the cause of ascites. In acute variceal bleed: IV Octreotide + IV Ceftriaxone + urgent Endoscopic Band Ligation (EVL) + restrictive transfusion (target Hb 7-8 g/dL).

	First-Line Medical Therapy	Definitive / Int
	IV Octreotide / Terlipressin + IV Ceftriaxone	Endoscopic Var
	Sodium restriction (<2g/d) + Spironolactone : Furosemide (100:40 ratio)	Therapeutic par
	IV Ceftriaxone or Cefotaxime (2g q8h x 5 days)	Long-term seco
ging	Lactulose (titrated to 2-3 soft stools/day) + Rifaximin (550mg BID)	Correct precipit